

INVERSE TRIGONOMETRY LEVEL 2

BASIC MCQs – 200 QUESTIONS

Pattern: Single-correct MCQs (like JEE/Boards mixed)

SECTION 1: BASICS – DEFINITIONS, DOMAIN, RANGE (Q1-Q25)

1. The principal value of $\sin^{-1} x$ lies in

A. $[0, \pi]$
B. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
C. $\left[0, \frac{\pi}{2}\right]$
D. $[-\pi, \pi]$

2. The principal value of $\cos^{-1} x$ lies in

A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
B. $[0, \pi]$
C. $[-\pi, \pi]$
D. $[0, 2\pi]$

3. The principal value of $\tan^{-1} x$ lies in

A. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
B. $[0, \pi]$
C. $\left(0, \frac{\pi}{2}\right)$
D. $(-\pi, \pi)$

4. The domain of $\sin^{-1} x$ is

A. $(-\infty, \infty)$
B. $[-1, 1]$
C. $(0, 1)$
D. $[0, 1]$

5. The range of $\sin^{-1} x$ is

A. $[0, \pi]$
B. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

- C. $[0, 2\pi]$
D. $[-\pi, \pi]$
6. The domain of $\cos^{-1} x$ is
A. $[-1, 1]$
B. $(-\infty, \infty)$
C. $[0, 1]$
D. $(0, 1)$
7. The range of $\cos^{-1} x$ is
A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
B. $[0, \pi]$
C. $[0, 2\pi]$
D. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
8. The domain of $\tan^{-1} x$ is
A. $[-1, 1]$
B. $(-\infty, \infty)$
C. $[0, \infty)$
D. $(0, \infty)$
9. The range of $\tan^{-1} x$ is
A. $[0, \pi]$
B. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
C. $[0, 2\pi]$
D. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
10. The domain of $\cot^{-1} x$ (principal value) is usually taken as
A. $(-\infty, \infty)$
B. $(0, \infty)$
C. $(-\infty, 0)$
D. $(-\infty, 0) \cup (0, \infty)$
11. The principal value range of $\cot^{-1} x$ is
A. $(0, \pi)$
B. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

C. $[0, \pi]$

D. $[-\pi, 0]$

12. The domain of $\sec^{-1} x$ is

A. $[-1, 1]$

B. $(-\infty, -1] \cup [1, \infty)$

C. $(-\infty, \infty)$

D. $[0, \infty)$

13. The range of $\sec^{-1} x$ (standard) is

A. $[0, \pi]$ excluding $\frac{\pi}{2}$

B. $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

C. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

D. $(0, \pi)$

14. The domain of $\csc^{-1} x$ is

A. $[-1, 1]$

B. $(-\infty, -1] \cup [1, \infty)$

C. $(-\infty, \infty)$

D. $[0, \infty)$

15. The range of $\csc^{-1} x$ (standard) is

A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ excluding 0

B. $\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$

C. $[0, \pi]$

D. $(0, \pi)$

16. $\sin^{-1} x$ is

A. Even function

B. Odd function

C. Neither even nor odd

D. Both even and odd

17. $\cos^{-1} x$ is

A. Even function

B. Odd function

- C. Neither even nor odd
- D. Both even and odd

18. $\tan^{-1} x$ is

- A. Even function
- B. Odd function
- C. Neither even nor odd
- D. Both even and odd

19. The function $\sin^{-1} x$ is

- A. Increasing on $[-1,1]$
- B. Decreasing on $[-1,1]$
- C. Constant on $[-1,1]$
- D. Increasing on $[0,1]$ and decreasing on $[-1,0]$

20. The function $\cos^{-1} x$ is

- A. Increasing on $[-1,1]$
- B. Decreasing on $[-1,1]$
- C. Constant on $[-1,1]$
- D. Increasing on $[0,1]$ and decreasing on $[-1,0]$

21. The function $\tan^{-1} x$ is

- A. Increasing on $(-\infty, \infty)$
- B. Decreasing on $(-\infty, \infty)$
- C. Constant
- D. Neither increasing nor decreasing

22. The value of $\sin^{-1} x$ is always

- A. Real for all x
- B. Real only for $|x| \leq 1$
- C. Imaginary for $|x| \leq 1$
- D. Imaginary for all x

23. If $y = \sin^{-1} x$, then x must satisfy

- A. $x \geq 0$
- B. $|x| \leq 1$
- C. $x > 1$
- D. $x < -1$

24. If $y = \cos^{-1} x$, then

- A. $|x| \leq 1$
- B. $|x| \geq 1$
- C. $x > 1$
- D. $x < -1$

25. If $y = \tan^{-1} x$, then

- A. $y \in [0, \pi]$
- B. $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- C. $y \in (-\pi, \pi)$
- D. $y \in [0, 2\pi]$

SECTION 2: DIRECT EVALUATION OF VALUES (Q26–Q55)

26. $\sin^{-1} 0$ is

- A. 0
- B. $\frac{\pi}{2}$
- C. π
- D. $-\frac{\pi}{2}$

27. $\sin^{-1} \left(\frac{1}{2}\right)$ equals

- A. $\frac{\pi}{2}$
- B. $\frac{\pi}{3}$
- C. $\frac{\pi}{6}$
- D. $\frac{2\pi}{3}$

28. $\sin^{-1} \left(\frac{\sqrt{3}}{2}\right) =$

- A. $\frac{\pi}{3}$
- B. $\frac{\pi}{6}$
- C. $\frac{\pi}{2}$
- D. $\frac{2\pi}{3}$

29. $\cos^{-1} 1 =$

A. π

B. 0

C. $\frac{\pi}{2}$

D. $-\frac{\pi}{2}$

30. $\cos^{-1} \left(\frac{1}{2} \right) =$

A. $\frac{\pi}{3}$

B. $\frac{\pi}{2}$

C. $\frac{\pi}{6}$

D. $\frac{2\pi}{3}$

31. $\cos^{-1} \left(-\frac{1}{2} \right) =$

A. $\frac{\pi}{3}$

B. $\frac{2\pi}{3}$

C. $\frac{5\pi}{6}$

D. $\frac{\pi}{6}$

32. $\tan^{-1} 1 =$

A. $\frac{\pi}{3}$

B. $\frac{\pi}{4}$

C. $\frac{\pi}{2}$

D. 0

33. $\tan^{-1} \sqrt{3} =$

A. $\frac{\pi}{6}$

B. $\frac{\pi}{3}$

C. $\frac{\pi}{4}$

D. $\frac{2\pi}{3}$

34. $\cot^{-1} 1$ (principal value) is

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. $\frac{3\pi}{4}$

D. $\frac{\pi}{3}$

35. $\sin^{-1} \left(-\frac{\sqrt{3}}{2} \right) =$

A. $-\frac{\pi}{3}$

B. $-\frac{\pi}{2}$

C. $\frac{\pi}{3}$

D. $\frac{\pi}{2}$

36. $\cos^{-1} 0 =$

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. π

D. 0

37. $\tan^{-1} 0 =$

A. 0

B. $\frac{\pi}{2}$

C. $-\frac{\pi}{2}$

D. π

38. $\sin^{-1} \left(\frac{\sqrt{2}}{2} \right) =$

A. $\frac{\pi}{6}$

B. $\frac{\pi}{4}$

C. $\frac{\pi}{3}$

D. $\frac{3\pi}{4}$

39. $\cos^{-1} \left(\frac{\sqrt{2}}{2} \right) =$

A. $\frac{\pi}{6}$

B. $\frac{\pi}{4}$

C. $\frac{\pi}{3}$

D. $\frac{3\pi}{4}$

40. $\tan^{-1}(-1)$ equals

A. $\frac{\pi}{4}$

B. $-\frac{\pi}{4}$

C. $\frac{3\pi}{4}$

D. $-\frac{3\pi}{4}$

41. $\sec^{-1} 2 =$

A. $\frac{\pi}{3}$

B. $\frac{\pi}{2}$

C. $\frac{\pi}{4}$

D. $\frac{2\pi}{3}$

42. $\csc^{-1} 2 =$

A. $\frac{\pi}{6}$

B. $\frac{\pi}{3}$

C. $\frac{\pi}{2}$

D. $\frac{2\pi}{3}$

43. $\cot^{-1} \sqrt{3}$ (principal value) equals

A. $\frac{\pi}{6}$

B. $\frac{\pi}{3}$

C. $\frac{\pi}{2}$

D. $\frac{2\pi}{3}$

44. $\sin^{-1}(-x)$ is equal to

A. $\sin^{-1} x$

B. $-\sin^{-1} x$

C. $\pi - \sin^{-1} x$

D. $-\pi - \sin^{-1} x$

45. $\tan^{-1}(-x)$ equals

A. $\tan^{-1} x$

B. $-\tan^{-1} x$

C. $\pi - \tan^{-1} x$

D. $-\pi - \tan^{-1} x$

46. $\cos^{-1}(-x)$ equals

A. $\cos^{-1} x$

B. $-\cos^{-1} x$

C. $\pi - \cos^{-1} x$

D. $\pi + \cos^{-1} x$

47. $\tan^{-1} 1 + \tan^{-1} 1 =$

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. π

D. 0

48. $\sin^{-1} x + \cos^{-1} x$ is equal to

A. 0

B. $\frac{\pi}{2}$

C. π

D. $\frac{3\pi}{2}$

49. $\tan^{-1} x + \cot^{-1} x$ equals

A. 0

B. $\frac{\pi}{2}$

C. π

D. $\frac{3\pi}{2}$

50. $\sin^{-1} \left(\frac{\sqrt{3}}{2} \right) + \cos^{-1} \left(\frac{1}{2} \right) =$

A. $\frac{\pi}{2}$

B. π

C. $\frac{2\pi}{3}$

D. $\frac{\pi}{3}$

51. $\sin^{-1} \left(-\frac{1}{2} \right) + \cos^{-1} \left(\frac{\sqrt{3}}{2} \right)$ equals

A. 0

B. $\frac{\pi}{2}$

C. π

D. $-\frac{\pi}{2}$

52. The value of $\tan^{-1} (\sqrt{3}) - \tan^{-1} (1)$ is

A. $\frac{\pi}{6}$

B. $\frac{\pi}{3}$

C. $\frac{\pi}{4}$

D. 0

53. The value of $\tan^{-1} (1) + \tan^{-1} (2)$ is

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. π

D. $\frac{3\pi}{4}$

54. The value of $\tan^{-1} (2) + \tan^{-1} (3)$ is

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. π

D. 0

55. $\sin^{-1} \left(\frac{3}{5} \right) + \cos^{-1} \left(\frac{4}{5} \right)$ is

A. 0

B. $\frac{\pi}{2}$

C. π

D. $\frac{3\pi}{2}$

SECTION 3: SIMPLE IDENTITIES & RELATIONS (Q56–Q90)

56. $\sin^{-1} x + \cos^{-1} x =$

- A. 0
- B. $\frac{\pi}{2}$
- C. π
- D. Depends on x

57. $\tan^{-1} x + \cot^{-1} x =$

- A. 0
- B. $\frac{\pi}{2}$
- C. π
- D. Depends on x

58. $\sin^{-1} x = \tan^{-1} \left(\frac{x}{\sqrt{1-x^2}} \right)$ is valid for

- A. $|x| > 1$
- B. $|x| < 1$
- C. $x \geq 0$ only
- D. $x \leq 0$ only

59. $\cos^{-1} x = \tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$ holds for

- A. $x < 0$
- B. $0 < x \leq 1$
- C. $|x| < 1$
- D. $|x| > 1$

60. $\sin^{-1} x$ can be written as $\cos^{-1} (?)$. The correct expression is

- A. $\cos^{-1} x$
- B. $\cos^{-1} (1 - x)$
- C. $\cos^{-1} \sqrt{1 - x^2}$
- D. $\cos^{-1} \sqrt{1 - x^2}$ with sign choice

61. $\sin^{-1} x + \sin^{-1} y = \sin^{-1} (x\sqrt{1-y^2} + y\sqrt{1-x^2})$ requires

- A. No condition
- B. $x^2 + y^2 \leq 1, xy \geq 0$
- C. $x^2 + y^2 \geq 1$
- D. $xy \leq 0$

62. If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$, then

- A. $x + y = 1$
- B. $x^2 + y^2 = 1$
- C. $x^2 + y^2 + 2xy = 1$
- D. $x + y = 0$

63. If $\sin^{-1} x = \cos^{-1} y$ and $x, y \in [0,1]$, then

- A. $x = y$
- B. $x + y = 1$
- C. $x^2 + y^2 = 1$
- D. $x^2 - y^2 = 0$

64. $\tan^{-1} \frac{1}{x} + \tan^{-1} x = \frac{\pi}{2}$ when

- A. $x < 0$
- B. $x > 0$
- C. $x = 0$
- D. Always

65. $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$ holds if

- A. $xy < 1$
- B. $xy > 1$
- C. $xy = 1$
- D. No condition

66. $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$ holds if

- A. $xy > -1$
- B. $xy < -1$
- C. $xy = 1$
- D. No condition

67. $\sin^{-1} x - \sin^{-1} y = \sin^{-1} (x\sqrt{1-y^2} - y\sqrt{1-x^2})$ requires

- A. No condition
- B. $x^2 + y^2 \leq 1$ and $x \geq y$
- C. $x^2 + y^2 \geq 1$
- D. $x, y > 1$

68. $\sin^{-1} x + \cos^{-1} y = \frac{\pi}{2}$ implies

A. $x = y$

B. $x^2 + y^2 = 1$ with $x, y \geq 0$

C. $x + y = 0$

D. $x = y = 0$

69. $\sin^{-1} x + \sin^{-1} \sqrt{1 - x^2} =$

A. 0

B. $\frac{\pi}{2}$ for $x \in [0, 1]$

C. π for all x

D. $\frac{3\pi}{2}$

70. $\cos^{-1} x + \cos^{-1} \sqrt{1 - x^2} =$ (for $x \in [0, 1]$)

A. 0

B. $\frac{\pi}{2}$

C. π

D. $\frac{3\pi}{2}$

71. If $x \in (0, 1)$, then $\tan^{-1} \left(\frac{2x}{1-x^2} \right) =$

A. $\tan^{-1} x$

B. $2\tan^{-1} x$

C. $\frac{\pi}{2} - \tan^{-1} x$

D. $\frac{\pi}{2}$

72. $\tan^{-1} x + \tan^{-1} \left(\frac{1-x}{1+x} \right)$ equals

A. 0

B. $\frac{\pi}{4}$

C. $\frac{\pi}{2}$

D. π

73. If $x^2 + y^2 + z^2 + 2xyz = 1$, then

A. $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = 0$

B. $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{\pi}{2}$

C. $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$

D. None

74. $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$ if

A. $\frac{x+y+z-xyz}{1-xy-yz-zx} = 0$

B. $xyz = 0$

C. $x + y + z = 0$

D. No condition

75. If $\sin^{-1} x = \tan^{-1} \frac{x}{\sqrt{1-x^2}}$, then $|x|$ must be

A. ≥ 1

B. ≤ 1

C. > 1

D. < 1

76. The expression $\cos^{-1} x + \cos^{-1} (-x)$ equals

A. 0

B. $\frac{\pi}{2}$

C. π

D. 2π

77. If $\sin^{-1} x = \cos^{-1} x$, then $\sin^{-1} x$ equals

A. 0

B. $\frac{\pi}{4}$

C. $\frac{\pi}{3}$

D. $\frac{\pi}{2}$

78. If $\tan^{-1} x + \tan^{-1} y = \frac{\pi}{4}$, then

A. $x + y = 1$

B. $\frac{x+y}{1-xy} = 1$

C. $\frac{x+y}{1-xy} = \tan \frac{\pi}{4} = 1$

D. All of the above relations are equivalent

79. If $\tan^{-1} a + \tan^{-1} b + \tan^{-1} c = \pi$, then

A. $a + b + c = abc$

- B. $a + b + c = 0$
- C. $ab + bc + ca = 0$
- D. None

80. $\sin^{-1} x + \cos^{-1} x$ is

- A. A constant function
- B. Increasing function
- C. Decreasing function
- D. Not defined

81. $\tan^{-1} x + \tan^{-1} \frac{1}{1-x}$ commonly appears in

- A. AP problems
- B. Telescoping sums with inverse tan
- C. Geometry only
- D. Complex numbers only

82. The relation $\sin^{-1} x + \sin^{-1} y = \cos^{-1} z$ represents

- A. Always a straight line
- B. Some relation between x, y, z on unit circle
- C. No geometric meaning
- D. Only inequality

83. If $\tan^{-1} x + \tan^{-1} y = \tan^{-1} z$, then generally

- A. $z = \frac{x+y}{1-xy}$ (under suitable conditions)
- B. $z = x + y$
- C. $z = xy$
- D. $z = 1 - xy$

84. For all real x , $\tan^{-1} x + \tan^{-1} (-x)$ equals

- A. 0
- B. $\frac{\pi}{2}$
- C. π
- D. $-\pi$

85. For $x > 0$, $\cot^{-1} x + \cot^{-1} \frac{1}{x} =$

- A. 0

B. $\frac{\pi}{2}$

C. π

D. 2π

86. If $\sin^{-1} x + \sin^{-1} y = \sin^{-1} (x + y)$, a necessary condition is

A. $x + y = 0$

B. $xy = 0$

C. $xy = 1$

D. $xy = 0$ and $x + y \in [-1, 1]$

87. $\cos^{-1} x - \cos^{-1} y = \cos^{-1} (xy + \sqrt{(1 - x^2)(1 - y^2)})$ involves

A. Sum identity

B. Difference identity

C. Product identity

D. No standard identity

88. If $x > 0$, $\tan^{-1} \frac{1}{x} =$

A. $\frac{\pi}{2} - \tan^{-1} x$

B. $\frac{\pi}{2} + \tan^{-1} x$

C. $-\tan^{-1} x$

D. $\tan^{-1} x$

89. The relation $\sin^{-1} x = \cos^{-1} \sqrt{1 - x^2}$ holds for

A. $x \in [-1, 1]$

B. $x \in [0, 1]$

C. $x \in [-1, 0]$

D. all real x

90. If $\tan^{-1} x + \tan^{-1} y = \tan^{-1} 1$, then $\frac{x+y}{1-xy} =$

A. 0

B. 1

C. -1

D. 2

SECTION 4: COMPOSITION WITH DIRECT TRIG FUNCTIONS (Q91–Q135)

91. $\sin(\sin^{-1} x)$ is equal to

- A. $|x|$
- B. x
- C. $-x$
- D. $\sqrt{1-x^2}$

92. $\cos(\cos^{-1} x)$ is equal to

- A. $|x|$
- B. x
- C. $-x$
- D. $\sqrt{1-x^2}$

93. $\tan(\tan^{-1} x)$ is equal to

- A. $|x|$
- B. x
- C. $-x$
- D. $1/x$

94. $\sin(\cos^{-1} x)$ equals

- A. x
- B. $\sqrt{1-x^2}$ (for suitable x)
- C. $-\sqrt{1-x^2}$
- D. $1/x$

95. $\cos(\sin^{-1} x)$ equals

- A. x
- B. $\sqrt{1-x^2}$ (for suitable x)
- C. $-\sqrt{1-x^2}$
- D. $1/x$

96. $\tan(\sin^{-1} x)$ is

- A. $\frac{x}{\sqrt{1-x^2}}$
- B. $\frac{\sqrt{1-x^2}}{x}$

C. $x\sqrt{1-x^2}$

D. $\frac{1-x^2}{x}$

97. $\sin(\tan^{-1} x)$ equals

A. $\frac{x}{\sqrt{1+x^2}}$

B. $\frac{\sqrt{1+x^2}}{x}$

C. $\frac{1}{\sqrt{1+x^2}}$

D. $\frac{1-x^2}{1+x^2}$

98. $\cos(\tan^{-1} x)$ equals

A. $\frac{x}{\sqrt{1+x^2}}$

B. $\frac{1}{\sqrt{1+x^2}}$

C. $\sqrt{1-x^2}$

D. $\frac{1-x^2}{1+x^2}$

99. If $\theta = \sin^{-1} \frac{3}{5}$, then $\cos \theta =$

A. $\frac{4}{5}$

B. $\frac{5}{3}$

C. $\frac{3}{4}$

D. $\frac{4}{3}$

100. If $\theta = \cos^{-1} \frac{4}{5}$, then $\sin \theta =$

A. $\frac{3}{5}$

B. $\frac{5}{4}$

C. $\frac{4}{3}$

D. $\frac{3}{4}$

101. If $x = \tan^{-1} \frac{3}{4}$, then $\sin x =$

A. $\frac{3}{5}$

B. $\frac{4}{5}$

C. $\frac{5}{3}$

D. $\frac{5}{4}$

102. If $x = \tan^{-1} \frac{3}{4}$, then $\cos x =$

A. $\frac{3}{5}$

B. $\frac{4}{5}$

C. $\frac{5}{3}$

D. $\frac{5}{4}$

103. If $\theta = \sin^{-1} \frac{5}{13}$, then $\sec \theta =$

A. $\frac{13}{12}$

B. $\frac{13}{5}$

C. $\frac{12}{5}$

D. $\frac{5}{12}$

104. If $\theta = \cos^{-1} \frac{5}{13}$, then $\csc \theta =$

A. $\frac{13}{12}$

B. $\frac{13}{5}$

C. $\frac{12}{5}$

D. $\frac{5}{12}$

105. $\sin^{-1}(\cos x)$ can be simplified using

A. Co-function identity and principal range

B. Only Pythagoras

C. Only symmetry

D. None

106. For $0 < x < \frac{\pi}{2}$, $\sin^{-1}(\cos x) =$

A. x

B. $\frac{\pi}{2} - x$

C. $\frac{\pi}{2} + x$

D. $\pi - x$

107. For $0 < x < \frac{\pi}{2}$, $\cos^{-1}(\sin x) =$

A. x

B. $\frac{\pi}{2} - x$

C. $\frac{\pi}{2} + x$

D. $\pi - x$

108. $\sin^{-1}(2\sin x \cos x)$ equals

A. $\sin^{-1}(\sin 2x)$

B. $2x$ always

C. $2x$ only if $2x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

D. None

109. $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ equals

A. $\sin^{-1} x$

B. $2\tan^{-1} x$ (with conditions)

C. $\cos^{-1} x$

D. $\pi - 2\tan^{-1} x$

110. $\cos^{-1}\left(\frac{1-t^2}{1+t^2}\right)$ equals

A. $2\tan^{-1} t$

B. $\tan^{-1} t$

C. $2\sin^{-1} \frac{2t}{1+t^2}$

D. $\sin^{-1} t$

111. $\sin^{-1}\left(\frac{2t}{1+t^2}\right)$ equals

A. $2\tan^{-1} t$ (for suitable t)

B. $\tan^{-1} t$

C. $\cos^{-1} t$

D. $\frac{\pi}{2} - 2\tan^{-1} t$

112. If $y = \sin^{-1}(2x - 1)$, then y lies in

A. $[0, \pi]$

B. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

C. $[0, \frac{\pi}{2}]$

D. $[-\pi, \pi]$

113. If $y = \cos^{-1}(2x - 1)$, then y lies in

A. $[0, \pi]$

B. $[-\pi, \pi]$

C. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

D. $[0, 2\pi]$

114. If $y = \tan^{-1}(2x)$, then

A. $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$

B. $y \in [0, \pi]$

C. $y \in [0, 2\pi]$

D. $y \in (-\pi, \pi)$

115. If $\theta = \sin^{-1} \frac{12}{13}$, then $\tan \theta =$

A. $\frac{5}{12}$

B. $\frac{12}{5}$

C. $\frac{13}{5}$

D. $\frac{5}{13}$

116. If $\phi = \tan^{-1} \frac{5}{12}$, then $\sec \phi =$

A. $\frac{13}{12}$

B. $\frac{13}{5}$

C. $\frac{12}{5}$

D. $\frac{5}{12}$

117. If $\phi = \tan^{-1} \frac{5}{12}$, then $\csc \phi =$

A. $\frac{13}{5}$

B. $\frac{13}{12}$

C. $\frac{5}{13}$

D. $\frac{12}{13}$

118. For $0 < x < 1$, $\sin(\sin^{-1} x + \cos^{-1} x) =$

- A. 0
- B. 1
- C. x
- D. $\sqrt{1 - x^2}$

119. For $0 < x < 1$, $\cos(\sin^{-1} x + \cos^{-1} x) =$

- A. 0
- B. 1
- C. x
- D. $\sqrt{1 - x^2}$

120. $\tan(\sin^{-1} x + \cos^{-1} x)$ equals

- A. 0
- B. 1
- C. ∞
- D. Undefined

121. If $y = \sin^{-1}(\cos x)$, then for $x \in [0, \pi]$, the range of y is

- A. $[0, \pi]$
- B. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- C. $\left[-\frac{\pi}{2}, 0\right]$
- D. $\left[0, \frac{\pi}{2}\right]$

122. If $y = \cos^{-1}(\sin x)$, with $x \in [0, \pi]$, the range of y is

- A. $[0, \pi]$
- B. $\left[0, \frac{\pi}{2}\right]$
- C. $\left[\frac{\pi}{2}, \pi\right]$
- D. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

123. $\cos(\tan^{-1} x) \cdot \sec(\tan^{-1} x)$ equals

- A. 1
- B. x
- C. $\sqrt{1 + x^2}$
- D. $\frac{1}{1 + x^2}$

124. $\sin(\tan^{-1} x) \cdot \csc(\tan^{-1} x)$ equals

- A. 1
- B. x
- C. $\sqrt{1+x^2}$
- D. $\frac{x}{\sqrt{1+x^2}}$

125. If $\alpha = \sin^{-1} \frac{3}{5}$ and $\beta = \cos^{-1} \frac{4}{5}$, then $\alpha + \beta =$

- A. 0
- B. $\frac{\pi}{2}$
- C. π
- D. 2π

126. The simplification of $\sin(\cos^{-1} \frac{4}{5} + \sin^{-1} \frac{3}{5})$ is

- A. 0
- B. 1
- C. $\frac{7}{25}$
- D. $\frac{24}{25}$

127. $\cos(\cos^{-1} a - \sin^{-1} b)$ (assuming $a^2 + b^2 = 1$) equals

- A. 0
- B. 1
- C. $a^2 - b^2$
- D. $2ab$

128. $\sin(\cos^{-1} a - \sin^{-1} b)$ (with $a^2 + b^2 = 1$) equals

- A. 0
- B. $a^2 - b^2$
- C. ab
- D. $a^2 + b^2$

129. If $\theta = \sin^{-1} x$, then $\cos(2\theta) =$

- A. $1 - 2x^2$
- B. $2x^2 - 1$
- C. $1 - x^2$
- D. $x^2 - 1$

130. If $\theta = \cos^{-1} x$, then $\cos (2\theta) =$

- A. $1 - 2x^2$
- B. $2x^2 - 1$
- C. $1 - x^2$
- D. $x^2 - 1$

131. $\tan (\sin^{-1} x) \cdot \tan (\cos^{-1} x)$ equals

- A. 1
- B. $\frac{x}{\sqrt{1-x^2}}$
- C. $\frac{x}{1-x^2}$
- D. $\frac{\sqrt{1-x^2}}{x}$

132. For $|x| \leq 1$, $\sin (\sin^{-1} x) \cdot \cos (\sin^{-1} x)$ equals

- A. $x\sqrt{1-x^2}$
- B. $1-x^2$
- C. x^2
- D. $\sqrt{1-x^2}$

133. If $\theta = \tan^{-1} x$, then $\sin (2\theta) =$

- A. $\frac{2x}{1+x^2}$
- B. $\frac{1-x^2}{1+x^2}$
- C. $\frac{2x}{1-x^2}$
- D. $\frac{1+x^2}{1-x^2}$

134. If $\theta = \tan^{-1} x$, then $\cos (2\theta) =$

- A. $\frac{1-x^2}{1+x^2}$
- B. $\frac{2x}{1+x^2}$
- C. $\frac{1+x^2}{1-x^2}$
- D. $\frac{2}{1+x^2}$

135. $\csc (\sin^{-1} x)$ equals

- A. $\frac{1}{x}$
- B. $\frac{1}{\sqrt{1-x^2}}$

- C. $\frac{x}{\sqrt{1-x^2}}$
D. $\sqrt{1-x^2}$

SECTION 5: BASIC EQUATIONS & RANGE/DOMAIN QUESTIONS (Q136–Q200)

136. The equation $\sin^{-1} x = \frac{\pi}{6}$ has solution
- A. $x = \frac{1}{2}$
B. $x = \frac{\sqrt{3}}{2}$
C. $x = \frac{\sqrt{2}}{2}$
D. $x = 1$
137. The equation $\cos^{-1} x = \frac{2\pi}{3}$ gives
- A. $x = -\frac{1}{2}$
B. $x = \frac{1}{2}$
C. $x = 0$
D. $x = \frac{\sqrt{3}}{2}$
138. The equation $\tan^{-1} x = \frac{\pi}{4}$ gives
- A. $x = \sqrt{3}$
B. $x = 1$
C. $x = \frac{1}{\sqrt{3}}$
D. $x = 0$
139. The equation $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ is satisfied by
- A. All $x \in [-1, 1]$
B. Only $x = 0$
C. Only $x = 1$
D. Only $x = -1$
140. The domain of $f(x) = \sin^{-1}(2x - 1)$ is
- A. $(-\infty, \infty)$
B. $[0, 1]$

C. $[0,1]$ shifted to satisfy $2x - 1 \in [-1,1]$

D. $[0,2]$

141. The domain of $f(x) = \cos^{-1}(2x^2 - 1)$ is

A. $(-\infty, \infty)$

B. $[-1,1]$

C. All real x since $|2x^2 - 1| \leq 1$ for all x

D. $|x| \leq 1$

142. The range of $y = \sin^{-1}(2x - 1)$ is

A. $[0, \pi]$

B. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

C. $[-\pi, \pi]$

D. Depends on x

143. The range of $y = \cos^{-1}(2x - 1)$ is

A. $[0, \pi]$

B. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

C. $[-\pi, \pi]$

D. $[0, 2\pi]$

144. The inequality $\sin^{-1} x \leq 0$ holds for

A. $x \in [0,1]$

B. $x \in [-1,0]$

C. $x \in (-\infty, 0]$

D. $x \in [-1,1]$

145. The inequality $\cos^{-1} x \geq \frac{\pi}{2}$ holds for

A. $x \in [0,1]$

B. $x \in [-1,0]$

C. $x \in [-1,1]$

D. $x \in [0, \infty)$

146. The inequality $\tan^{-1} x \geq 0$ holds for

A. $x \geq 0$

B. $x > 0$ only

C. $x \leq 0$

D. All x

147. The solution of $\sin^{-1} x = \sin^{-1} \frac{3}{5}$ is

A. $x = \frac{3}{5}$ only (principal branch)

B. $x = \frac{3}{5}$ or other values

C. Infinite solutions in reals

D. No solution

148. The solution of $\cos^{-1} x = \cos^{-1} \frac{4}{5}$ is

A. $x = \frac{4}{5}$ only

B. $x = -\frac{4}{5}$ only

C. $x = \pm \frac{4}{5}$

D. No solution

149. The solution of $\tan^{-1} x = \tan^{-1} \frac{7}{3}$ is

A. $x = \frac{7}{3}$ (principal)

B. $x = \frac{7}{3} + k\pi$

C. $x = \tan \left(\tan^{-1} \frac{7}{3} + k\pi \right)$

D. Infinite but principal is $\frac{7}{3}$

150. The domain of $f(x) = \tan^{-1} \left(\frac{1-x^2}{1+x^2} \right)$ is

A. All real x

B. $|x| \neq 1$

C. $|x| < 1$

D. $|x| > 1$

151. The range of $f(x) = \tan^{-1} \left(\frac{1-x^2}{1+x^2} \right)$ is

A. $[0, \pi]$

B. $\left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$

C. $(-\pi, \pi)$

D. $[0, 2\pi]$

152. The range of $f(x) = \sin^{-1}(\sin x)$ (principal) is

- A. $[0, \pi]$
- B. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- C. $(-\pi, \pi)$
- D. $[-\pi, 0]$

153. The range of $f(x) = \cos^{-1}(\cos x)$ is

- A. $[0, \pi]$
- B. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- C. $[\pi, 2\pi]$
- D. $(-\infty, \infty)$

154. If $f(x) = \sin^{-1}(x^2)$, then domain is

- A. $(-\infty, \infty)$
- B. $|x| \leq 1$
- C. $|x| \geq 1$
- D. $x \in [-1, 1]$ (actually same as B)

155. If $g(x) = \cos^{-1}(2x^2 - 3)$, domain is defined by

- A. $|2x^2 - 3| \leq 1$
- B. $|2x^2 - 3| \geq 1$
- C. $2x^2 - 3 \geq 1$
- D. $2x^2 - 3 \leq -1$

156. The function $f(x) = \tan^{-1}(x^2)$ has domain

- A. $(-\infty, \infty)$
- B. $|x| \leq 1$
- C. $|x| \geq 1$
- D. $[0, \infty)$

157. $\sin^{-1} x$ is

- A. Onto from $[-1, 1]$ to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- B. Many-one from $[-1, 1]$ to same range
- C. Not invertible
- D. Both A and B

158. $\cos^{-1} x$ is

- A. Decreasing and one-one on $[-1,1]$
- B. Increasing and one-one
- C. Constant
- D. Not invertible

159. $\tan^{-1} x$ is a bijection from

- A. $\mathbb{R} \rightarrow (0, \pi)$
- B. $\mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- C. $(0, \pi) \rightarrow \mathbb{R}$
- D. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$

160. The inequality $\sin^{-1} x < \frac{\pi}{4}$ implies

- A. $x < \frac{\sqrt{2}}{2}$
- B. $x > \frac{\sqrt{2}}{2}$
- C. $x \leq 0$
- D. No solution

161. The inequality $\cos^{-1} x > \frac{\pi}{3}$ gives

- A. $x < \frac{1}{2}$
- B. $x > \frac{1}{2}$
- C. $x \leq -\frac{1}{2}$
- D. $x \geq -\frac{1}{2}$

162. The inequality $\tan^{-1} x < \frac{\pi}{6}$ gives

- A. $x < \frac{1}{\sqrt{3}}$
- B. $x > \frac{1}{\sqrt{3}}$
- C. $x = \frac{1}{\sqrt{3}}$
- D. No solution

163. If $\sin^{-1} x = \cos^{-1} x$, then

- A. $x = 0$
- B. $x = \frac{1}{\sqrt{2}}$

C. $x = \frac{1}{2}$

D. $x = -\frac{1}{2}$

164. The number of solutions of $\sin^{-1} x = \frac{\pi}{3}$ in $[-1,1]$ is

A. 0

B. 1

C. 2

D. Infinite

165. The number of solutions of $\cos^{-1} x = \frac{\pi}{3}$ in $[-1,1]$ is

A. 0

B. 1

C. 2

D. Infinite

166. The number of solutions of $\tan^{-1} x = \frac{\pi}{3}$ in $\mathbb{R}$ is

A. 0

B. 1 (principal)

C. Infinite

D. 2

167. If $\sin^{-1} x + \sin^{-1} x = \frac{\pi}{2}$, then $x =$

A. 0

B. $\frac{\sqrt{2}}{2}$

C. 1

D. $-\frac{\sqrt{2}}{2}$

168. If $\cos^{-1} x + \cos^{-1} x = \pi$, then $x =$

A. 0

B. $\frac{1}{2}$

C. $-\frac{1}{2}$

D. 1

169. The domain of $f(x) = \sin^{-1} (3x - 2)$ is

A. $\frac{1}{3} \leq x \leq 1$

B. $0 \leq x \leq 1$

C. $\frac{1}{3} \leq x \leq \frac{5}{3}$

D. $-\infty < x < \infty$

170. The range of $f(x) = \sin^{-1}(3x - 2)$ (on its domain) is

A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

B. $[-\pi, \pi]$

C. $[0, \pi]$

D. $\left[-\frac{\pi}{2}, 0\right]$

171. If $y = \cos^{-1}(1 - 2x^2)$, then y equals

A. $2\sin^{-1} x$ (for $|x| \leq 1$)

B. $\sin^{-1} x$

C. $2\tan^{-1} x$

D. $\tan^{-1} x$

172. The function $y = \tan^{-1}(\sin x)$ has range

A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

B. $[-\tan^{-1} 1, \tan^{-1} 1]$

C. $[-1, 1]$

D. $[0, \pi]$

173. The function $y = \sin^{-1}(\tan x)$ is defined for those x where

A. $|\tan x| \leq 1$

B. $|\tan x| \geq 1$

C. $x \in \mathbb{R}$

D. $x \in (0, \pi)$

174. The equation $\sin^{-1} x = \cos^{-1}(1 - 2x^2)$ is equivalent to

A. $x = \frac{1}{2}$

B. $x^2 = \frac{1}{2}$

C. $\sin^{-1} x = 2\sin^{-1} x$

D. none

175. If $f(x) = \tan^{-1}(x) + \tan^{-1}\left(\frac{1-x}{1+x}\right)$, then $f(x)$ is

A. Constant

- B. Increasing
- C. Decreasing
- D. Not defined

176. For $x > 0$, the inequality $\tan^{-1} x < \frac{\pi}{4}$ means

- A. $x < 1$
- B. $x > 1$
- C. $x = 1$
- D. No solution

177. The inequality $\sin^{-1} x > \cos^{-1} x$ in $[-1, 1]$ holds for

- A. No x
- B. All x
- C. $x > \frac{1}{\sqrt{2}}$
- D. $x < \frac{1}{\sqrt{2}}$

178. The equation $\sin^{-1} x + \sin^{-1} (1 - x) = \frac{\pi}{2}$ holds for

- A. All $x \in [0, 1]$
- B. Only $x = \frac{1}{2}$
- C. $x = 0$ or 1
- D. None

179. The domain of $f(x) = \cos^{-1} (x^2 - 2)$ is defined by

- A. $x^2 - 2 \in [-1, 1]$
- B. $x^2 - 2 \in (-\infty, \infty)$
- C. $x^2 - 2 \geq 1$
- D. $x^2 - 2 \leq -1$

180. The function $f(x) = \tan^{-1} (x) + \tan^{-1} (-x)$ is identically

- A. $\frac{\pi}{2}$
- B. 0
- C. π
- D. $-\pi$

181. The value of $\sin^{-1} x$ at $x = 1$ is

- A. 0

B. $\frac{\pi}{2}$

C. π

D. not defined

182. The value of $\cos^{-1} x$ at $x = -1$ is

A. 0

B. $\frac{\pi}{2}$

C. π

D. not defined

183. The value of $\tan^{-1} x$ as $x \rightarrow \infty$ tends to

A. 0

B. $\frac{\pi}{2}$

C. $-\frac{\pi}{2}$

D. π

184. The value of $\tan^{-1} x$ as $x \rightarrow -\infty$ tends to

A. 0

B. $\frac{\pi}{2}$

C. $-\frac{\pi}{2}$

D. $-\pi$

185. The graph of $y = \sin^{-1} x$ is obtained from $y = \sin x$ by

A. Reflection in x-axis

B. Reflection in line $y = x$ with restricted domain

C. Translation

D. Rotation

186. The graph of $y = \cos^{-1} x$ is obtained from $y = \cos x$ by

A. Reflection in x-axis

B. Reflection in line $y = x$ and domain restriction

C. Vertical stretch

D. Horizontal shift only

187. Among the following, which is NOT invertible on $[-1,1]$?

A. $\sin x$ restricted to $[-\frac{\pi}{2}, \frac{\pi}{2}]$

- B. $\cos x$ restricted to $[0, \pi]$
- C. $\tan x$ restricted to $(-\frac{\pi}{2}, \frac{\pi}{2})$
- D. $\sin x$ restricted to $[0, \pi]$

188. The function $f(x) = \sin^{-1} x$ is continuous on

- A. $[-1, 1]$
- B. Entire $\mathbb{R}$
- C. $(0, 1)$ only
- D. $[-1, 0) \cup (0, 1]$

189. The function $f(x) = \cos^{-1} x$ is continuous on

- A. $[-1, 1]$
- B. $(-\infty, \infty)$
- C. $[0, \infty)$
- D. none

190. The function $f(x) = \tan^{-1} x$ is continuous on

- A. $\mathbb{R}$
- B. $\mathbb{R} \setminus \{0\}$
- C. $(-\infty, \infty)$ except at $\pm \frac{\pi}{2}$
- D. None

191. The derivative sign of $\sin^{-1} x$ (without computing formula) suggests the function is

- A. Increasing on $[-1, 1]$
- B. Decreasing on $[-1, 1]$
- C. Constant
- D. Not monotonic

192. $\cos^{-1} x$ is differentiable on

- A. $(-1, 1)$
- B. $[-1, 1]$
- C. $(-\infty, \infty)$
- D. None

193. The inverse of $y = \sin^{-1} x$ is

- A. $y = \sin x$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- B. $y = \cos x$

- C. $y = \tan x$
- D. Not defined

194. The inverse of $y = \tan^{-1} x$ is

- A. $y = \tan x$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- B. $y = \tan x$ on $(0, \pi)$
- C. $y = \sin x$
- D. $y = \cos x$

195. Among these, whose inverse is multi-valued unless a range is fixed?

- A. $\sin x$
- B. $\cos x$
- C. $\tan x$
- D. All of these

196. The composition $\sin(\cos^{-1} x)$ is defined for

- A. All real x
- B. $|x| \leq 1$
- C. $|x| \geq 1$
- D. $x > 0$

197. The composition $\cos(\sin^{-1} x)$ is defined for

- A. All real x
- B. $|x| \leq 1$
- C. $|x| \geq 1$
- D. $x > 0$

198. The composition $\tan(\cos^{-1} x)$ is defined for

- A. $|x| \leq 1$
- B. $|x| < 1$
- C. $x > 0$
- D. Only $x = 0$

199. The equation $\sin^{-1} x = \cos^{-1} x$ has

- A. No solution
- B. Exactly one solution in $[-1, 1]$
- C. Two solutions
- D. Infinitely many

200. The total number of MCQs in this LEVEL 2 set is

- A. 100
- B. 150
- C. 200
- D. 250